

1. CORE INSPECTION METADATA

Timestamp	2026-09-23T22:10:16.725916
Primary Finding	DENT
Severity Layer	HIGH
Detections Logged	1
Analysis Engine	YOLOv11-Nano (Local Edge)

2. ACQUIRED IMAGERY & BOUNDING BOXES



Fig 1. AI-annotated structural imagery overlay.

3. RAW DETECTION COORDINATES

ID	DEFECT CLASS	CONFIDENCE	SEVERITY	BOUNDING BOX [X, Y, W, H]
01	DENT	86.0%	HIGH	[0.430, 0.418, 0.310, 0.207]

4. RAG CONTEXT RETRIEVAL

Query N/A

Source Manual ac_43.13-1b_w-chg1.pdf (Page 150)

Extracted Snippet: *"sustain these stresses, and then allow them to return into the structure. The repair must be equal to the original structure, but not stronger or stiffer, which will cause stress concentrations or alter the resonant frequency of the structure. c. All-metal aircraft are made of very thin sheet metal, and it is possible to restore the strength of the skin without restoring its rigidity. All repairs should be made using the same type and thickness of material that was..."*

5. SYNTHESIZED REPAIR PROTOCOL

Generated via Local Phi-3 Mini strictly grounded in retrieved maintenance documentation.

1 Isolate the damaged area

Clearly mark and isolate the high-severity dent on the aircraft's surface to ensure focused attention during repair.

2 Remove the dent

Carefully remove the dented metal using appropriate tools, ensuring not to cause additional damage to the surrounding area.

3 Prepare the surface

Clean and prepare the dented area by removing any debris, rust, or paint, creating a smooth, bare metal surface for repair.

4 Apply the repair material

Select the same type and thickness of metal used in the original construction and apply it to the dented area, ensuring a seamless integration with the surrounding skin.

5 Shape and finish the repair

Shape the applied material to match the contour of the surrounding skin, and finish by polishing to restore the aircraft's original appearance and aerodynamic properties.